WEEK 3 · PRACTICE SESSION · 2 HOURS · TEAMS OF 3–4

DNS, HTTP and a reverse proxy, hands-on

Mode: Computer · **Portfolio item:** request-flow diagram

Goal: See with real tools what happens between "type a URL" and "page loads".

Before the session (students)

Install the lab environment (Node.js 22, Docker Desktop, Git; see Practice Sessions, "Lab environment"). Test with `docker run hello-world`.

Part A — DNS (25 min)

1. Look up records and note the TTL:

```
dig www.google.com A
dig www.google.com AAAA
dig gmail.com MX
dig +trace rupp.edu.kh          # follow root → .kh → authoritative
```

(Windows without dig: `nslookup -type=A www.google.com`.)

2. Run `dig www.google.com` twice and compare the TTL values. Explain why the second one is smaller.
3. Find one domain that returns **several A records**. What is this used for?

Part B — HTTP and TLS (30 min)

1. Run:

```
curl -v https://example.com -o /dev/null
curl -I --http2 https://www.cloudflare.com
curl -I --http3 https://www.cloudflare.com   # if your curl supports HTTP/3
```

Identify in the output: the IP and port connected to, the TLS version, the HTTP version, the status code and 3 response headers.

2. Open browser DevTools → Network on any news site. Record: number of requests, total size, time to first byte of the HTML, and the protocol column (h2/h3).
3. **CDN check:** `curl -I` a static image or JS file from a big site twice. Look for `cache-control`, `age`, `cf-cache-status` or `x-cache`. Was it a cache HIT?

Part C — Reverse proxy with nginx (45 min)

1. Create a folder `week3/` with these files:

`app/index.html`

```
<h1>FoodHub backend</h1>
```

`app/server.mjs`: a tiny Node web server that logs who calls it

```
import { createServer } from 'node:http';
import { readFileSync } from 'node:fs';

createServer((req, res) => {
```

```
  console.log(`${req.method} ${req.url} from ${req.socket.remoteAddress}, X-Forwarded-For:
${req.headers['x-forwarded-for']}`);
  res.writeHead(200, { 'Content-Type': 'text/html' });
  res.end(readFileSync('/srv/index.html'));
}).listen(3000, () => console.log('app listening on 3000'));
```

`nginx.conf`

```
events {}
http {
  server {
    listen 80;
    location / {
      proxy_pass http://app:3000;
      proxy_set_header X-Forwarded-For $remote_addr;
      add_header X-Served-By "foodhub-proxy";
    }
  }
}
```

`docker-compose.yml`

```
services:
  app:
    image: node:22-alpine
    volumes: ["./app:/srv"]
    command: node /srv/server.mjs
  proxy:
    image: nginx:1.27
    volumes: ["./nginx.conf:/etc/nginx/nginx.conf:ro"]
    ports: ["8080:80"]
    depends_on: [app]
```

2. `docker compose up`, then `curl -i http://localhost:8080`. Find the `X-Served-By` header. Note that port 3000 is **not** reachable from your laptop (`curl localhost:3000` fails): only the proxy is exposed.
3. Read the app logs (`docker compose logs app`). Which IP does the app see as the client, and which IP is in `X-Forwarded-For`? Why are they different?
4. Stop the app container (`docker compose stop app`) and curl again. Which status code does nginx return? (Expected: 502 Bad Gateway.)

Part D — Request-flow diagram (20 min)

Draw the full path of a customer opening the FoodHub menu page: phone → DNS resolver → CDN (for images) → reverse proxy → backend → database. Mark where TLS ends and where caching happens.

Before you leave: checklist

- ☐ DNS answers recorded with TTL explanation
- ☐ HTTP/TLS details identified from curl output; CDN hit/miss observed
- ☐ nginx proxy running; answers to questions 3 and 4 of Part C
- ☐ Request-flow diagram in the portfolio

Record your results

Part A — DNS

Query	Answer (IP / host)	TTL first run → second run
www.google.com A		
www.google.com AAAA		
gmail.com MX		
dig +trace rupp.edu.kh: the three levels		

Why is the second TTL smaller? / Domain with several A records and what it is used for:

Part B — HTTP and TLS

Item	What you found
IP and port connected to	
TLS version	
HTTP version (example.com / cloudflare h2 / h3)	
Status code	
Three response headers	
DevTools: requests, total size, TTFB, protocols	
CDN check: headers seen, HIT or MISS?	

Part C — Reverse proxy

Question	Answer
Value of X-Served-By	
Result of curl localhost:3000, and why	
Client IP the app sees vs X-Forwarded-For, and why they differ	
Status code when the app is stopped	

Part D — Request-flow diagram

Draw the diagram on A3 or in draw.io and attach it. Mark where TLS ends and where caching happens.